MIL-P-2067D
30 September 1968
SUPERSEDING
MIL-P-2067C
21 March 1967

MILITARY SPECIFICATION

PAILS, METAL, MEASURING (FOR BAKERIES)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers pails for measuring liquids and flour.

1.2 Classification.- The measuring pails shall be of the following types as specified (see 6.2).

Type I - Liquid measuring, 4-gallon capacity, graduated in gallons.

Type II - Flour measuring, 40-pound capacity.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-250/2	-	Aluminum Alloy 3003, Plate and Sheet
QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon
QQ-W-461	-	Wire, Steel, Carbon, (Round, Bare and Coated)
PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Box, Fiberboard
PPP-B-640	-	Box, Fiberboard, Corrugated, Triple Wall, 350 Pound Maximum Weight

FSC 7240

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PPP-C-650 - Crates, Wood, Open and Covered
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-T-10727 - Tin Plating; Electrodeposited or Hot-Dipped, For Ferrous and Nonferrous Metals

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MS 51931 - Rivet, Solid-Tinner's

DRAWING

ARMY NATICK LABORATORIES

5-11-272 - Pails, Metal, Measuring for Bakeries

(Miniature copy of Drawing 5-11-272 identified as figure 1 is attached for information purposes only.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample measuring pail shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

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* 3.2 Material.- Materials not definitely specified shall be of the quality normally use by the manufacturer in his standard commercial practice.

3.2.1 Steel, sheet and strip.- Steel sheet shall conform to cold rolled, commercial quality and steel strip shall conform to cold rolled, half-hard temper of QQ-S-698.

3.2.2 Wire steel.- Steel wire shall conform to steel number 1010, 1015, or 1020, finish 3 of QQ-W-461.

* 3.2.3 Aluminum, sheet and strip.- Aluminum alloy sheet and strip shall conform to 3003 of QQ-A-250/2.

* 3.2.4 Rivets.- Tinner's rivets shall conform to MS 51931.

* 3.3 Design and construction.- The pails shall be designed and constructed as specified herein and on Drawing 5-11-272, except the type I pail may be of the alternate design specified in 3.4. Nonmetallic sealing compound shall not be used in the construction of steel pails. The pails shall not leak when tested as specified in 4.4.1.

3.3.1 Side, bottom, and ears.- The side, bottom, and ears shall be formed from one piece of material specified in 3.2.1.

3.3.2 Top edge.- The top edge of the pail shall be beaded over the reinforcing wire specified in 3.2.2 to form a tightly rolled bead.

3.3.3 Bails.- The bails shall be fabricated from wire specified in 3.2.2 and shall be similar to those shown on Drawing 5-11-272.

* 3.4 Type I, alternate design, aluminum.- The type I, alternate design pail shall have a minimum capacity of 4 gallons. The pail body shall be formed from one piece of aluminum alloy sheet specified in 3.2.3 with tapered sides. The bail shall be as specified in 3.3.3. The ears shall be aluminum and shall be of the offset type so designed to permit the bail to swing freely from one side of the pail to the other. The ears shall be riveted to the outside of the pail at points diametrically opposite each other. Each ear shall be attached by a minimum of two aluminum rivets.

* 3.4.1 Graduation marks.- The aluminum pail shall have graduated markings indicating one gallon increments. The graduations shall be marked on the inside surface and shall extend around the circumference of the pail. The markings shall be free of any sharp corners and shall be clear and legible from a distance of 3 feet. The volume shall not vary more than plus or minus 5 percent at any graduation when tested as specified in 4.4.3.

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* 3.5 Marking for identification.- Identification markings shall be on the outside bottom of the pails. Markings shall include the manufacturer's name and trademark and otherwise shall conform to the manufacturer's standard identification marking.

3.6 Finish.-

* 3.6.1 Type I and II, steel pails.- The completely fabricated steel pails, without bail attached, shall be cleaned, and loose scale, rust, and other foreign matter removed. All surfaces of the pails, inside and outside, and the bail shall be completely coated with tin in accordance with type I or II of MIL-T-10727. The thickness of the coating shall be a minimum 0.0003 inches when tested as specified in 4.4.2. The coating shall be free from lumps, blisters, gritty areas, uncoated spots, acid, black spots, dross, and flux.

* 3.6.2 Aluminum pail.- The type I aluminum pail shall have all rough and sharp edges, dirt, grease, or foreign matter removed in accordance with the manufacturer's standard practice. The pail body and ears shall be finished with a smooth commercial, standard bright finish and shall be uniform in appearance. The bail shall be cleaned and tin coated as specified in 3.6.1.

3.7 Workmanship.- The pails shall be free of cracks, dents, malformation, or deep scratches. All burrs, splinters, and sharp edges shall be removed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required it shall be examined for the defects listed in table I and dimensions specified herein and tested as specified in 4.4.

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4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.2 End item inspection.- The inspection lot shall consist of all pails of one type offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated pail.

4.3.2.1 Visual examination.- The end item shall be examined for defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish (general)	Finish not as specified.	X	
	Not finished where required.	X	
Steel pails	Flaking, peeling, blistered, rust, scale, areas of no coating, lumps, acid, black spots, dross and flux.	X	
	Tear drops or dripping on edges.	X	
	Dirt, grit, or other foreign matter embedded in finish.	X	
	Scratches through coating.	X	
Aluminum pail	Not a smooth bright finish.	X	
Design for steel pails	Nonmetallic sealing compound used.	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship	Any component dented, creased or malformed.	X	
	Any burrs, slivers, splinters, or sharp edges.	X	
Body, steel pails (as applicable)	Not joined by lock seam.	X	
	Top edge of pail not beaded over reinforcing wire.	X	
Aluminum pails	Not one piece construction.	X	
	Not tapered sides.	X	
Bottom, steel pails	Not lock seamed and crimped to sides.	X	
Ears	Ears do not permit the loops of bail to swing freely.	X	
	Not formed as specified.	X	
	Not attached as specified.	X	
	Missing.	X	
Graduation marks for type I	Missing.	X	
Bail	Missing.	X	
Riveting	Rivet missing.	X	
	Rivet loose.	X	
Marking for identification	Omitted, missing, incomplete, or not applied in location specified.		X

4.3.2.2 Dimensional examination.- Inspection shall be made of the sample unit to determine compliance with dimensions specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be level S-2 and the AQL shall be 4.0 defects, expressed in terms of defects per hundred units.

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4.3.3 Testing of the end product.- The end item shall be tested as specified in 4.4.1, 4.4.2 and 4.4.3, when applicable. The inspection level shall be S-2 and the AQL shall be 4.0 defects expressed in terms of defects per hundred units.

4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packing and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

TABLE II.- Preparation for delivery defects

Examination	Defect
Marking, exterior	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing, damaged, or otherwise defective.

4.3.4.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-601, PPP-B-621, PPP-B-636, PPP-B-640, or PPP-C-650, examination for defects in closure, waterproofing, and strapping shall be in accordance with the applicable container specification.

4.4 Tests.-

* 4.4.1 Leakage.- Pails shall be filled with water and allowed to stand for 5 minutes. The pails shall then be examined to determine compliance with 3.3.

4.4.2 Tin coating.- Tin coating thickness shall be determined by the test procedure specified in MIL-T-10727 to determine compliance with 3.6.1.

4.4.3 Capacity test.- The capacity of the type I aluminum pail shall be tested by measuring the volume of water at each graduated marking on the pail. The volumes shall be checked to determine compliance with 3.4.1.

5. PREPARATION FOR DELIVERY

5.1 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.-

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5.1.1.1 Type I and II steel pails.- Six pails of one type shall be packed three in length, two in width, and one in depth in a snug-fitting crate conforming to type I, style A of PPP-C-650. Alternatively, shipping containers may be used that conform to overseas type of PPP-B-601 or class 2, style 2 or 4 of PPP-B-621. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

5.1.1.2 Type I aluminum pail.- Six pails shall be nested in the most compact manner and packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. The pails shall be positioned with the bail ears in the corners of the container. Fiberboard inserts shall be used to fill voids in each corner of the container. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.1.2 Level B.-

5.1.2.1 Type I and II steel pails.- Six pails of one type shall be packed three in length, two in width, and one in depth in a snug-fitting crate conforming to type I, style B of PPP-C-650. Alternatively shipping containers may be used that conform to domestic type, style A or B of PPP-B-601; class 1, style 2 or 4 of PPP-B-621; or class 1, style E of PPP-B-640.

5.1.2.2 Type I aluminum pail.- Six pails shall be nested in the most compact manner and packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. The pails shall be positioned with the bail ears in the corners of the container. Fiberboard inserts shall be used to fill voids in each corner of the container. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

5.1.2.2.1 When specified (see 6.2) the shipping container shall be V3c or V3s, fabricated and closed in accordance with PPP-B-636. The shipping container material may also be grade V4s of PPP-F-320.

5.1.3 Level C.- The pails shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.2 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.- The pails are intended for measuring flour and water for bread making. The pails are components of the Baking Plant, Field Portable, MIL-B-43322.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) First article.- When a preproduction sample is not required (see 3.1).
(d) Selection of applicable level of packing (see 5.1).
(e) When weather-resistant fiberboard shipping containers are required for level B packing (see 5.1.2.2.1).

6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the pervious issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - GL
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Preparing activity:

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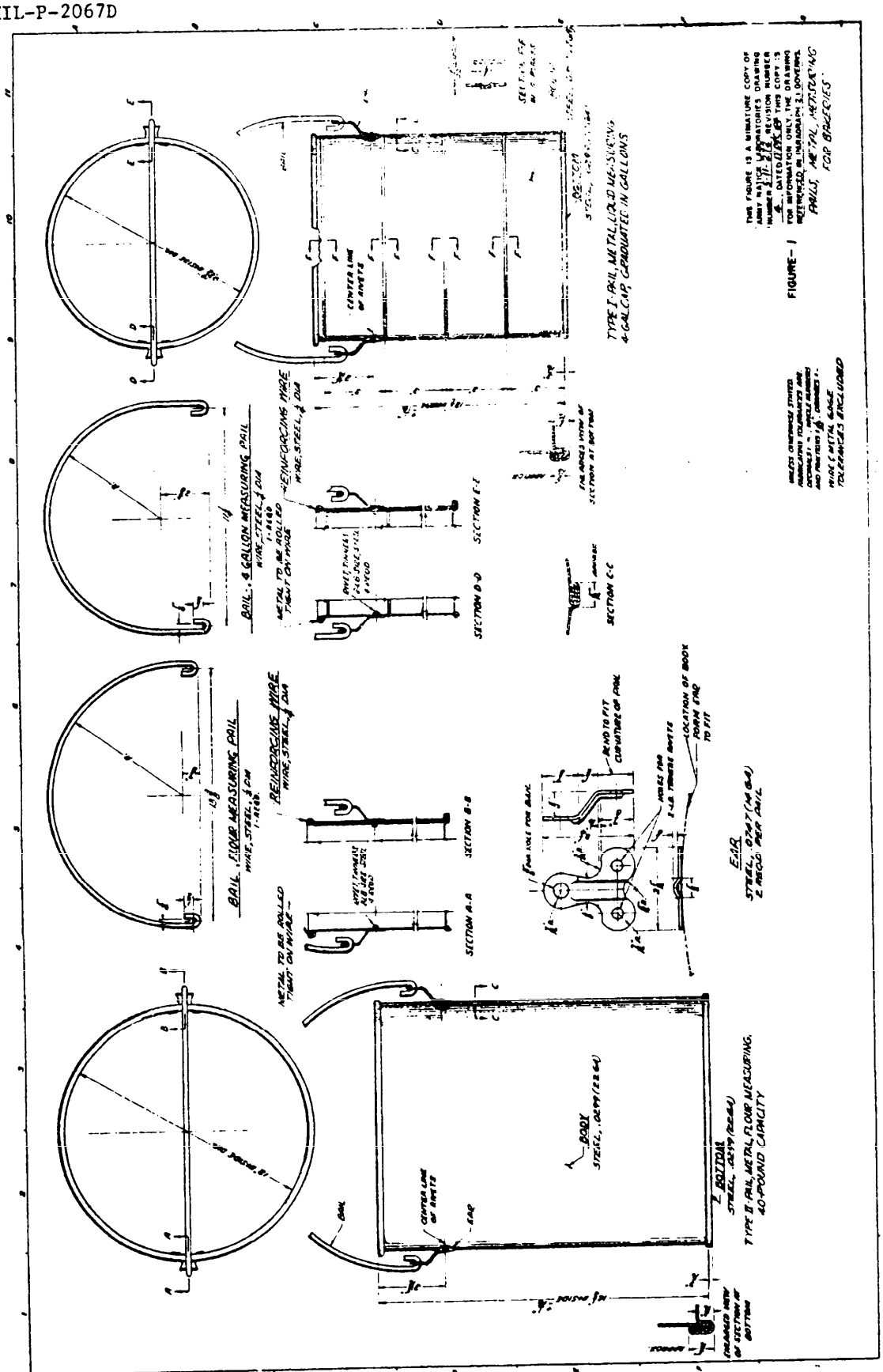


FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

RAILS, METAL, MEASURING (FOR [illegible]), MIL-P-[illegible]

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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